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REVIEW

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Adverse effects of antibiotics in children with cancer: are short-course antibiotics for febrile neutropenia part of the solution?

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ABSTRACT

Introduction: Febrile neutropenia is a common complication experienced by children with cancer or those undergoing hematopoietic stem cell transplantation. Repeated episodes of febrile neutropenia result in cumulative exposure to broad-spectrum antibiotics with potential for a range of serious adverse effects. Short-course antibiotics, even in patients with high-risk febrile neutropenia, may offer a solution.

Areas covered: This review addresses the known broad effects of antibiotics, highlights developments in understanding the relationship between cancer, antibiotics, and the gut microbiome, and discusses emerging evidence regarding long-term adverse antibiotic effects. The authors consider available evidence to guide the duration of empiric antibiotics in pediatric febrile neutropenia and directions for future research.

Expert opinion: Broad-spectrum antibiotics are associated with antimicrobial resistance, *Clostridioides difficile* infection, invasive candidiasis, significant disturbance of the gut microbiome and may seriously impact outcomes in children with cancer or undergoing allogenic hematopoietic stem cell transplant. Short-course empiric antibiotics are likely safe in most children with febrile neutropenia and present a valuable opportunity to reduce the risks of antibiotic exposure.

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1. Introduction

There have been remarkable improvements in the treatment and outcomes for most childhood cancers. The five-year survival for children with acute lymphoblastic leukemia (ALL) now approaches 90% in most high-income countries [1]. This reflects not just advancements in hematopoietic stem cell transplant (HSCT), chemo- and immuno-therapeutics but a collaborative focus on improving supportive care. Early identification of febrile neutropenia (FN) and rapid administration of broad-spectrum of antibiotics has underpinned the management strategy. However, antimicrobial resistance (AMR) is a looming risk, threatening to stall and even reverse improvements in mortality [2]. While antimicrobial stewardship efforts in the oncology population are often driven by the risk of AMR, the complications and unintended consequences of antibiotic use are in fact much broader.

Antibiotic exposure is cumulative across a patient's cancer journey, mostly driven by repeated empiric treatment of FN. Fever in association with cancer was reported as early as 1845 and remains one of the most common complications of cancer treatment [3,4]. The risk of serious bacterial infection is directly proportional to the depth and duration of neutropenia. Intensive phases of chemotherapy during treatment of pediatric acute leukemia and lymphoma, as well as conditioning for HSCT, result in long periods of neutropenia and greater risk of serious bacterial infection [5]. In these patient groups, 50% of neutropenic episodes are associated with fever [6]. The standard-of-care is giving immediate broad-spectrum antibiotics at the onset of fever and time-to-antibiotic has become a key outcome measure of sepsis programs worldwide [7,8]. Despite dramatic advancements in microbiological and molecular diagnostics, and excellent outcomes for children, the duration of antibiotics in those without proven infection remains poorly defined and international FN guidelines are conflicting. Short course antibiotics for FN are potentially a high-value target for reducing cumulative antibiotic exposure. This review will summarize recommendations for antibiotic duration in pediatric FN and evaluate emerging evidence of the impact of antibiotics on the human microbiome, response to cancer therapy, and the risk of graft-versus-host disease (GVHD).

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Article highlights

- Intravenous antibiotic therapy in children with cancer is associated with antimicrobial resistance, *C difficile* infection, invasive candidiasis, decreased quality of life and is costly.
- Antibiotics in children with cancer are associated with disruption of the gut microbiome, which may impact response to cancer therapy and long-term outcomes.
- In children undergoing allogenic hematopoietic stem cell transplantation, there is a complex interaction between antibiotics, the microbiome and immune homeostasis. This may impact transplant-specific outcomes, including survival.
- Short-course antibiotics in select children with high-risk febrile neutropenia and no identified infection are likely safe and represent an opportunity to reduce antibiotic exposure, however further clinical trials are needed.

2. Current recommendations and evidence for empiric antibiotic duration in children with febrile neutropenia

Early recognition of FN and prompt antibiotic administration is universally endorsed and a benchmark of supportive care [7]. Fortunately, the majority of FN episodes in children are not associated with microbiologically documented infection or sepsis and the absolute risk of serious outcomes (death or intensive care admission) is approximately 3% in high-income countries [6,9–11]. The corollary is that most antibiotics are administered empirically for FN in children who ultimately have fever of unknown source (5). While longer courses of antibiotics are often required for clinically or microbiologically diagnosed infection, the recommended duration of empiric antibiotics for fever or unknown source is variable and, not surprisingly, there are significant differences in practice [12–14].

Many guidelines now recommend early cessation of antibiotics in children at low-risk of bacterial infection (low-risk FN) who have negative blood cultures, are afebrile for more than 24–48 hours and are clinically stable (Table 1). The identification of low-risk FN has been extensively evaluated, with the derivation of over 25 pediatric FN clinical decision rules (CDRs) worldwide [5,15]. These CDR's reliably predict children at low-risk of infection or adverse outcome and, when used in combination with home-based care pathways, have been shown to safely reduce hospital length of stay, decrease costs of care, and improve quality of life [16–21].

In children with high-risk FN (expected absolute neutrophil count <0.5 cells/mm^3 for ≥7 days), the default approach is to continue antibiotics until absolute neutrophil count (ANC) recovery. Given the median duration of neutropenia is 11 days in children with chemotherapy-associated FN and 18 days in children undergoing induction treatment for ALL, this translates to many additional days of antibiotic exposure [6]. The benefits of ceasing antibiotics earlier are balanced against the potential risks of partially treated invasive infection, physiological instability, and recurrence of fever. Unfortunately, there is a paucity of randomized control trial (RCT) level data addressing the question of optimal antibiotic duration in this population. Antibacterial prophylaxis is used by some centers to reduce the risk of FN events in higher-risk patients with conflicting data on impact on antimicrobial resistance, C. difficile infection, and other adverse events [26,27]. The efficacy of recommencing prophylaxis following an episode of FN is also poorly understood. To date, biomarker guided strategies have not been recommended in pediatric or adult FN guidelines. Although a number of studies have explored their predictive performance, heterogeneity in study design and outcomes, conflicting results, cost, and availability limit their routine use [28]. An updated systematic review, incorporating data from 41 studies with 8319 FN episodes, found procalcitonin and interleukin 6- and 8- better predicted infection or adverse outcome compared with CRP [28]. PANTHER Cub [29], a recent pilot RCT in children (n = 16 FN episodes) found that procalcitonin-guided antibiotic therapy for FN is likely safe and could reduce antibiotic duration and spectrum. Novel diagnostics such as next-generation sequencing and blood transcriptomics also show promise and may facilitate early detection of bacterial infection [30,31]. Ultimately, the role of biomarkers in FN care pathways and any impact on antibiotic duration is yet to be defined.

There are nine RCTs that compare prolonged (i.e. continuation of antibiotics until recovery of ANC) with short-course (i.e. discontinuation of antibiotics irrespective of ANC) antibiotic therapy in adults with FN [18]. Just two have been conducted within the last two decades and only one study focused on the high-risk patient population (acute leukemia or HSCT) and this study excluded children [32]. Furthermore, no FN RCT has been adequately powered to assess the safety of short-course antibiotics. Despite these limitations, short-course antibiotics have been reported to reduce antibiotic exposure without an increase in adverse events or bacterial infection [18]. In total, just 399 episodes of FN in children have been enrolled in randomized clinical trials evaluating duration of therapy (Table 2) and only one trial included high-risk FN episodes (88/399 episodes) [33–36]. This study found that in patients with FN and confirmed viral respiratory infection, stopping antibiotics at 48 h was not associated with higher rates of clinical failure compared to continuing antibiotics for at least 7 days [34]. Due to the lack of RCT-level data in pediatric high-risk FN, a recent Cochrane Collaboration review concluded that the duration of antibiotics for children with sustained bone marrow suppression is unknown [18]. Antibiotic duration for children with high-risk FN has been identified as a critical evidence gap by international expert clinicians and researchers [16]. The management of FN is also a research priority for families who highlight the frequency of FN episodes, variation in care and physical, psychological, and social impacts of FN [37]. Minimizing cumulative exposure to antibiotics across episodes of FN could be a key pillar of anti-microbial stewardship and address the broad adverse effects of antibiotics in children with cancer. The next challenge is two-fold; implementation of short-course antibiotics consistently in low-risk FN and safely optimizing duration of antibiotics in the high-risk FN group.

3. Negative impacts of antibiotics in children with cancer or undergoing hematopoietic stem cell transplantation

Antibiotics, while mostly well tolerated, may have a range of adverse effects such as nausea, vomiting, nephrotoxicity,

Table 1. Current international recommended duration of empiric antibiotic therapy for febrile neutropenia.

Guideline (Publication year)	Recommended duration of antibiotics for unexplained fever
UK National Institute of Health and Care Excellence [22] (2020)	Continue until "response to treatment" (irrespective of ANC)
Eighth European Conference on Infections in Leukemia [23] (2020)	Low-risk and selected high-risk patients: consider de-escalation (to oral antibiotics or cease) after ≥ 72 h if hemodynamically stable and afebrile for 24–48 h, regardless of ANC or expected duration of neutropenia
International Pediatric Fever and Neutropenia Guideline [16] (2017)	Low risk: consider discontinuation of empirical antibiotics at 72 h in patients who have negative blood cultures and who have been afebrile for ≥ 24 h, irrespective of marrow recovery status, as long as careful follow-up is ensured All other patients: discontinue empirical antibiotics in patients who have negative blood cultures at 48 h, who have been afebrile for ≥ 24 h, and who have evidence of marrow recovery
European Society for Medical Oncology [24] (2016)	Low-risk: Continue until afebrile for 5–7d (or ANC ≥ 500 cells/mm^3) High-risk: Continue for up to 10d (or until ANC ≥500 cells/mm^3)
Infectious Diseases Society of America [25] (2010)	Continue until the patient is afebrile for at least 48 h and ANC ≥500 cells/mm^3

h, hours; d, days; ANC, absolute neutrophil count.

electrolyte disturbance, hypersensitivity reactions, and added myelosuppression [38–43]. In children without cancer, the negative impacts of antibiotics are well described. Antibiotics are the most common cause of adverse drug reactions in children and are estimated to be responsible for almost 70,000 emergency department visits annually in the United States (US) [42,43]. Antibiotics are also associated with long-term adverse health effects including allergic conditions, asthma, juvenile idiopathic arthritis, increased weight gain, and obesity [44]. In addition to these potential effects, children with cancer are vulnerable to broader effects; AMR, disruption of the microbiome, increased rates of invasive candidiasis and infection with *Clostridioides difficile* [45–57]. Inpatient therapy for FN is also costly, with an estimated average cost per child of $16,326 to $20,396 (Australian dollars, AUD) [20,58]. Likewise, admission to hospital for antibiotics has implications for individual and family quality of life, including measurable disruption to family life, appetite, and sleep quality [19,59].

3.1. Antibiotic-resistant infection

The World Health Organization identifies AMR as a 'significant global threat of far-reaching proportions that requires an urgent comprehensive multisectoral approach [60]. In 2019, there were an estimated 4.95 million global deaths attributable to AMR [61]. Given prior antibiotic use and severe neutropenia are independent risk factors for AMR, children with cancer are at risk of bearing the burden of AMR [45,62]. It is increasingly recognized that AMR complicates care for children with cancer and threatens to curb improvements in mortality [2,63]. AMR in those with cancer and undergoing HSCT is associated with poor outcome, including prolonged fever, longer duration of bacteremia, increased risk of admission to ICU and higher mortality [2,45–47,62,64].

Prior or prolonged antibiotic exposure is consistently shown to contribute to development of AMR in international studies. Other risk factors include prolonged hospital admission, previous colonization or infection with AMR bacteria and dose-intense chemotherapy [45,46]. In a multinational observational study (15 centers, eight countries) of 1,291 bloodstream infections in 1031 children with cancer or undergoing HSCT, between 22% and 30% of Gram-negative infections and up to 58% of clinically significant Gram-positive infections were resistant to one or more first-line antibiotics for FN. Antibiotic exposure, especially carbapenem use, was associated with resistant Gram-negative bloodstream infection and mortality [45]. In India, the rates of AMR in children with cancer and Gram-negative bloodstream infection are more alarming with extended-spectrum beta-lactamase (ESBL)-producing, carbapenem-resistant, or multidrug-resistant organisms reported in 24–59%, 10–27%, and 44% of cases, respectively [62]. Similarly, a retrospective study in Israel of 270 children showed that Gram-negative bloodstream infection was common, occurring in 16.3% with hematological malignancy and 29.5% following HSCT, with 13.4–25.8% of isolates testing multidrug resistant and 10% carbapenem-resistant. Carbapenem-resistant, compared with carbapenem-sensitive bloodstream infection, were associated with a 2.1 day longer mean duration of bacteremia, a higher risk for admission to intensive care (44.4% vs 10.1%), and higher mortality rate (33% vs 5.8%) [47].

Antimicrobial resistance in pediatric HSCT recipients is of particular concern due to the increased prevalence of Gram-negative infection, the challenges of treatment, and the high mortality associated with infection [47]. Potential drivers of AMR in HSCT are prolonged hospitalization, longer and more severe neutropenia, gastrointestinal GVHD causing compromise of gut barrier function and cumulative exposure to antibiotics [62,65]. While there is no international surveillance in this population, there is a signal of increasing Gram-negative resistance. A multicentre US study in pediatric allo-HSCT (2004–2012) reported resistance to a first-line antibiotic (piperacillin-tazobactam or ceftazidime) in 21% and a carbapenem in 8% [65]. In a recent international prospective study (2014–2015) that included 79 children (<18y) post-HSCT, carbapenem resistance occurred in 26.6% and multidrug resistance in 40.4% of Gram-negative infections [66].

Some of the factors contributing to AMR in the susceptible cancer population are modifiable and this is reflected in variable rates of AMR (and mortality) between centers. In particular, strategies to reduce antibiotic exposure have been associated with lower rates of AMR in immunocompromised children [67]. More broadly, there is a critical need for AMR surveillance at a local and international level as the foundation

Table 2. Summary of pediatric randomized controlled trials investigating short-course antibiotics for febrile neutropenia.

Study (country & year)	Patient population and inclusion criteria	Intervention & comparator	Primary outcome	Key results (intervention v comparator)	Limitations
Kumar [36] (India, 2020)	75 FN episodes • Leukemia/Lymphoma 62 (83%) • Solid tumor 13 (17%) Low-risk FN (malignancy in remission and expected neutropenia <10 d), antibiotics commenced in outpatient setting	Stop antibiotics after afebrile > 24 h and negative blood culture Change to oral antibiotics (amoxicillin-clavulanate and levofloxacin) until ANC ≥ 500 cells/mm^3	Clinical success: afebrile and no readmission during same period of neutropenia	• Clinical success 36/38 (95%) vs 35/37 (95%) (p = 0.682) • No readmissions in either arm • No positive BC post randomization in either arm	• Target sample size was 142 (stopped early due to slow recruitment) • Not blinded
Klaassen [35] (Canada, 2000)	73 FN episodes • Leukemia 48 (66%) • Lymphoma 5 (7%) • Brain tumor 7 (9%) • Other: 13 (18%) Low-risk FN (remission with no comorbidities)	Oral placebo after afebrile >24 h, negative BC and no clinical signs of sepsis Continue oral antibiotics (cloxacillin and cefixime) for 14d or until ANC ≥ 500 cells/mm^3	Clinical failure (any of): positive BC, readmission, death or recurrence of fever during same period of neutropenia	• Clinical failure 2/36 patients (5.6%) vs 5/37 patients (13.5%) (p = 0.43) • High rates of side effect reported in antibiotic arm (31% vs 11%; p = 0.095), mostly gastrointestinal symptoms • 1/36 (2.8%) in the placebo arm had a positive BC • No deaths • Average costs of care $1,753 ($934–$10,404 CAD) vs $1,887 ($951–$7012)	• Only included 27% of all FN episodes • Inadequately powered • Majority with bone marrow recovery at time of discharge
Santolaya [34] (Chile, 2017)	176 FN episodes • Leukemia/Lymphoma 104 (59%) • Relapsed leukemia 11 (6%) • Solid tumor 61 (35%)	Stop empiric antibiotics after > 48 h hospitalization, clinically stable (CRP < 90 mg/L, "favourable clinical evolution," no positive bacterial culture) and NP sample positive for a respiratory virus Continue IV ceftriaxone (LR) or ceftazidime + amikacin (HR) for 7d, afebrile for ≥24 h and CRP < 40 mg/L	Uneventful resolution (any of): no positive BC, change or re-installment of antibiotic therapy, recurrence of fever, hemodynamic instability, or death during same period of neutropenia	• Uneventful resolution 80/84 (95%) vs 89/92 (97%) (OR 1.48; 95% CI 0.32–6.83, p = 0.61) • Significant reduction in antibiotic duration (median 3 vs 7d) • No deaths. • 1 episode of sepsis requiring ICU admission in comparator arm • No difference in LOS, days of fever or demonstrated/probable invasive bacterial infection	• Included only if afebrile/clinically stable at 48 h • Excluded HSCT • CRP-driven approach • Not blinded • Only 75% had respiratory symptoms at time of collecting NP sample
Santolaya [33] (Chile, 1997)	75 FN episodes • Leukemia 33 (44%) • Lymphoma 3 (4%) • Solid tumor 39 (52%)	Stop empiric antibiotics after >48 h hospitalization and clinically stable (CRP < 40 mg/L on day 1 or 2, no clinical instability and no documented infection) Continue anti-staphylococcal penicillin and third-generation cephalosporin or aminoglycoside until neutropenia and fever resolved	Clinical failure (any of): positive BC, clinically diagnosed bacterial infection, recurrence of fever, hemodynamic instability, 2 consecutive measurements of CRP > 40 or death during same period of neutropenia	• Clinical failure 2/36 (5.6%) vs 3/39 (7.7%) • No episodes of sepsis, ICU admission or death in either arm • No difference in LOS or adverse events (inc. bacterial infection)	• Small study, single site • Excluded HSCT • Fever defined as two measurements ≥38.7°C separated by 1 h • Biomarker driven approach

FN, febrile neutropenia; h, hours; ANC, absolute neutrophil count; vs, versus; ICU, intensive care unit; LOS, length of stay; CRP, C-reactive protein; NP, nasopharyngeal; LR, low risk; HR, high risk; BC, blood culture; OR, odds ratio; d, days; CAD, Canadian dollars; HSCT, hematopoetic stem cell transplant; IV, intravenous.

to identify risk factors for AMR, inform antimicrobial stewardship and benchmark antimicrobial use.

3.2. Invasive fungal infection

Invasive fungal infection (IFI) is a critical complication of cancer therapy, given its high morbidity and mortality [68–71]. Antibiotic exposure disrupts normal aerobic intestinal flora, increases colonization with *Candida* spp., and may alter the host response to fungal infection [72]. An animal model shows that exposure to antibiotics, especially vancomycin, primes for invasive candidiasis by damaging gut barrier function and impairing gastrointestinal antifungal immunity [73]. Numerous studies show an association between prolonged or broad-spectrum antibiotics and invasive candidiasis in neonates and children with and without malignancy [52,74–77]. In children admitted to the ICU (including 17/101 with malignancy), administration of vancomycin (OR 6.2) or agents with anerobic activity (OR 3.5) for greater than three days was significantly associated with candidemia [75].

Three studies have specifically examined the association of antibiotic use with IFI in children with cancer [52–54]. In one retrospective cohort study, children with acute myeloid leukemia complicated by proven or probable IFI had a median exposure to broad-spectrum antibiotics that was double that of children without IFI (23 vs 11 days) and duration of broad-spectrum antibiotics was an independent risk factor for IFI (OR 2.04, $P < 0.01$) [52]. Likewise, children with leukemia complicated by *Candida* spp. meningitis had mean antibiotic exposure 5.5 days longer compared with children with no meningitis (12 vs 6.5 days, OR 1.25, 95% CI = 1.03–1.51) [53]. In a retrospective study of 334 children with malignancy, IFI only occurred in those with administration of antibiotics for greater than one week [54]. Beyond interventions to limit unnecessary antibiotic exposure in this population, anti-fungal prophylaxis should be considered in patients that require prolonged courses of antibiotics and who have additional risk factors for invasive candidiasis [78].

3.3. Clostridioides *difficile infection*

Clostridioides difficile is a spore forming Gram-positive organism mostly associated with diarrhea but capable of a broad range of severe complications including dehydration, abdominal perforation, sepsis, and death [79]. *C difficile* infection (CDI) occurs through a series of events; disruption of the healthy intestinal microbiome, fecal-oral transmission of *C difficile* spores, colonization of the gastrointestinal tract, dominance of a toxigenic strain and production of *C difficile* toxins A and B [80,81]. CDI is common in children with cancer, occurring at a rate 15 times that of hospitalized children without cancer [82], and 10% of cases are complicated by severe colitis, ileus, or disease requiring surgery [83]. Children with cancer are especially susceptible to CDI due to broad-spectrum antibiotic exposure, immunosuppression, prolonged hospitalization, gastric acid suppression, and parenteral feeding [82,84]. In children with cancer or post-HSCT, CDI is associated with prolonged hospitalization, higher health-care costs, and increased risk of death [79,82], and recurrent disease occurs in 10–25% of cases [83,84].

The association between antibiotic exposure and CDI in children is well established and antibiotic use is the most important modifiable risk factor in children with cancer [55,56]. The risk of developing CDI is proportional to exposure to broad-spectrum antibiotics, such as clindamycin, third-generation cephalosporins, fluoroquinolones, and carbapenems [55,85,86]. In a cohort of 8268 children with newly diagnosed ALL, 268 (3.2%) developed CDI within 180 days of diagnosis and every day of anti-pseudomonal β-lactam use was associated with an increased risk of CDI (hazard ratio, 1.05; 95% confidence interval, 1.01–1.09) [57]. Antimicrobial stewardship interventions targeting prolonged exposure to select antibiotics, namely piperacillin-tazobactam, cephalosporins, and fluoroquinolones, are capable of reducing rates of CDI and may impact duration of hospital-stay and mortality [87,88].

3.4. Disruption to the microbiome

The gut microbiome (GM) actively regulates immune and inflammatory tone, including the host response to cancer, efficacy of cancer therapy, and the risk of cancer-associated complications [89,90]. A multistep model of childhood leukemia proposes that the GM is a factor driving transformation of pre-leukemic clones to cancer cells in predisposed individuals [91]. This is supported by pre-clinical models showing that bacteria may either promote or suppress malignant cell growth. In one study, mice treated with antibiotics or raised in germ-free conditions exhibited defective host anti-tumor response, rapid tumor progression, and dampened response to chemotherapy while mice subject to complete gut sterilization have suppressed tumor growth [92–94]. In another mouse sarcoma model, treatment with vancomycin causes abnormal T-helper 17 cellular response and renders cyclophosphamide chemotherapy less effective [95]. Taken together, these animal models suggest a complex interplay between the host's intestinal flora and immune defenses, including against cancer.

In children with cancer, both chemotherapy and antibiotics are associated with perturbation of the GM; however, the impact on treatment course, survival, and long-term clinical outcomes is less clear. In a case-control study comparing stool microbiome in patients with ALL and healthy siblings at varying time points over one year, microbial diversity was significantly lower at diagnosis in those with ALL and further decreased with antibiotic exposure [48]. In a longitudinal study of 199 children with newly diagnosed ALL, GM diversity decreased significantly in the intensive phases of chemotherapy (induction and re-induction) but was not clearly associated with antibiotic exposure. Microbial diversity recovered post-consolidation therapy and antibiotic exposure; however, microbial composition was abnormal. Relative abundance of Proteobacteria prior to chemotherapy and emergence of Enterococcaceae as a dominant organism during chemotherapy independently predicted the risk of FN [49]. This is concordant with several other studies showing that dysbiosis, marked by organism predominance or reduction in microbial diversity, is associated with increased risk of infectious complications [50,51], and gut dysbiosis leading into HSCT may predict the risk of bloodstream infection [96–99]. In the largest of these studies, during HSCT there were significant shifts in the GM, where dominance of *Enterococcus* spp. increased the risk of enterococcal bacteremia while dominance of Proteobacteria (anaerobic Gram-negatives including Enterobacteriaceae) increased the risk of Gram-negative bacteremia [96].

It is hypothesized that gut dysbiosis negatively impacts the host immune response to malignancy [100]. Furthermore, antibiotic-induced disturbance of the GM directly undermines the efficacy of immune checkpoint inhibitors (ICIs), a targeted cancer immunotherapy. Reduction in GM diversity and dominance of pathobionts in those receiving ICIs are associated with significantly worse progression-free and overall survival [101]. As the use of ICIs in children expands, concurrent exposure to antibiotics must be considered and adjusted for in future pediatric clinical trials [102].

Mucositis is a common complication of cancer therapy and results in treatment disruption, bacteremia, added resource utilization and reduced quality of life [103,104]. At the gut level, the microbiome regulates mucosal immune response and use of antibiotics may disrupt the oral-gut microbiome axis and play a crucial role in the pathogenesis of mucositis [105]. In a mouse model, there is GM dysbiosis and injury to the intestinal mucosa within 96 hours of cisplatin exposure; however, administration of healthy microbiota supports intestinal epithelial recovery and prevents bacterial translocation from the gut [106]. In a prospective study of children with newly diagnosed ALL and healthy siblings, stool microbial diversity was lower at diagnosis and decreased with induction therapy, with peak overgrowth of stool *Enterococcus species* around day 15 of induction, correlating with peak systemic inflammation and serum citrulline (a marker of mucositis) [107]. Enteral nutrition may protect the GM and mucosal integrity to reduce translocation of gut bacteria. In 42 consecutive children undergoing HSCT, enteral nutrition was associated with reduced incidence of bloodstream infection compared with parenteral nutrition [108]. Probiotics also show promise in restoring the GM and reducing both oral-mucositis and chemotherapy-induced diarrhea. In a meta-analysis of 1,013 adult patients with malignancy, probiotics were associated with a significant reduction in oral mucositis, however included trials were small, often non-randomized, used a variety of single and multi-strain probiotics, and excluded children. Therefore, robust clinical trials are needed to understand the oral-gut microbiome and establish the type, dose, and efficacy of probiotics in the management of mucositis in pediatric cancer patients [109].

Children and young people who survive cancer are at increased risk of a range of long-term health consequences. Obesity, metabolic syndrome, and cardiovascular disease, all conditions also linked with antibiotic exposure, are a particular challenge [44,110]. In the St. Jude Lifetime Cohort, 40% of adult survivors of childhood leukemia followed up for more than 10 years were obese (body mass index ≥30 kg/m^2) and 33.6% were diagnosed with metabolic syndrome [111,112]. Given diversity and richness of the GM is shown to be impacted for at least 1-year after completing therapy for ALL, antibiotic exposure during cancer therapy has the potential to influence long-term health outcomes in children surviving cancer [48]. Further studies are needed to describe microbiota dynamics in survivors and the potential implications this may have for broad health outcomes.

3.5. Graft-versus-host disease (GVHD)

In the HSCT setting, the GM plays a crucial immunomodulatory role and appears to influence immune reconstitution, the risk of GVHD and transplant-related mortality. By the time a child proceeds to HSCT, often after multiple cycles of chemotherapy and episodes of FN, the GM has changed significantly [48–51,96–99]. Furthermore, during HSCT, the GM remains dynamic. Morjaria *et al*. [113] performed high resolution daily stool sampling in 18 adult patients undergoing HSCT and showed marked shifts in bacterial density and microbiota composition day-to-day, with the greatest shift in response to antibiotic exposure. Specifically, piperacillin-tazobactam and meropenem, even more than metronidazole, resulted in near complete loss of obligate anaerobes. Similarly, in children undergoing allo-HSCT, alpha-diversity, a measure of microbiota diversity and richness, was 2.4-fold lower pre-HSCT, compared with healthy controls, and decreased further in the gut, nose, and mouth to week + 3 [114].

In patients undergoing HSCT, there is an established association between antibiotic exposure, dysbiosis, and life-threatening complications [115]. In the largest study, Peled *et al*. [116] used 16s ribosomal RNA sequencing to characterize the GM in 1,362 adult patients undergoing allo-HSCT and found that loss of intestinal alpha diversity during transplant was associated with overall transplant-related mortality and death due to acute graft-versus-host disease (aGVHD). Seven studies in children detail the risk of antibiotic exposure and aGVHD (Table 3) [117–123]. In 2550 patients aged 1–21y (age <16 y, n = 2014) undergoing allo-HSCT, carbapenems, antipseudomonal penicillins, and cephalosporins given in the early transplant period (start of conditioning to day +7) were associated with grade 2–4 aGVHD. Specifically, in adjusted models, carbapenem use (particularly prior to day 0) was a significant risk factor for gastrointestinal and high-grade aGVHD [119] .

Both pediatric and adult evidence supports an antibiotic class effect in which systemic antibiotics with broader anaerobic activity are associated with disturbance of the GM and greater risk of aGVHD. In a single center retrospective cohort study that included 607 children, antibiotics with anaerobic activity given for FN in the early HSCT period (Day −7 to Day +28) increased the risk of gut or liver aGVHD and were associated with an increased risk of GVHD-related death [120]. In a study of 15 children undergoing allo-HSCT, cumulative days of therapy with anaerobic antibiotics were associated with development of grade 2–4 aGVHD and gastrointestinal GVHD [122]. Romick-Rosendale *et al*. [121] performed 16S metagenomic sequencing and metabolomics in 42 children undergoing allo-HCT, and showed that anaerobic antibiotic exposure resulted in loss of GM diversity, decreased short-chain fatty acids (SCFAs) at day +14 and simultaneous reduction in fecal butyrate levels associated with an increased risk of aGVHD.

The specific association of anti-anaerobic antibiotics with acute GVHD is congruent with immunological models of GVHD

Table 3. Pediatric studies investigating antibiotic exposure and graft-versus-host disease (GVHD).

Author (country & year)	Study type	Population	Antibiotics investigated (HSCT exposure period)	Graft versus host disease data	Comments	Level of evidence*
Gardner [117] (USA, 2022)	Single center retrospective cohort study	443 allo-HSCT in children and adults aged from 6 m to 39y (median age 6.7y)	Levofloxacin prophylaxis (n = 227) (Day −2 to ANC ≥200 cells/ mm^3)	• Grade 2–4 aGVHD lower in prophylaxis vs no prophylaxis group (4% vs 12%, p = 0.01) • Mean total antibiotic days lower in prophylaxis vs no prophylaxis group (35d vs 47d, p = 0.001)	Pediatric data not reported separately 49 (22%) in prophylaxis group received abatacept for GVHD prophylaxis 21 (9%) in prophylaxis group had CDI compared with 42 (20%) in the no prophylaxis group in the first 100d post-HSCT (*P* = 0.003)	Level 2b
Galazka [118] (Poland, 2021)	Multicentre retrospective cohort study	459 consecutive children undergoing allo-HSCT across all transplant centers	Oral decontamination with any of: colistin (n = 231), ciprofloxacin (n = 164), gentamicin (n = 93), metronidazole (n = 86), cotrimoxazole (n = 52), rifaximin (n = 17) or cefuroxime (n = 14) (conditioning to ANC recovery)	• Prophylactic gut decontamination decreased risk of GI aGVHD (HR 0.5, 95% CI 0.2 – 0.8, *P* = 0.014) • Gentamicin associated with lower rate of aGVHD II–IV (13% vs 44%), aGVHD III–IV (5% vs 22%), and GI aGVHD (3% vs 22%) compared with other/no antibiotics (*P* < 0.001) • Ciprofloxacin associated with higher rate of aGVHD II–IV (53% vs 30%), aGVHD III–IV (35% vs 12%) and GI aGVHD (28% vs 13%) compared with other/no antibiotics (*P* < 0.001) • Colistin associated with higher rate of aGVHD II–IV (51% vs 26%), aGVHD III–IV (30% vs 8%) and GI aGVHD (28% vs 8%) compared with other/no antibiotics (*P* < 0.001)	203 (44%) colonized by AMR bacteria (rectal swabs). Colonization with AMR bacteria was associated with GI GVHD. Patients received >1antibiotic as decontamination and a variety of IV antibiotics during neutropenia.	Level 2b
Elgarten [119] (USA, 2021)	Multicentre retrospective cohort (CIBMTR registry data)	2550 allo-HSCT patients (age 1–21y) with acute leukemia across 36 US centers (2014/ 2550 age < 16y)	Broad-spectrum cephalosporins, anti-pseudomonal penicillins and carbapenems (conditioning to Day +7)	*Primary outcome: Grade 2–4 aGVHD* • Cephalosporins vs none: aOR 1.05 (95% CI 0.83–1.32) • Anti-pseudomonal penicillin vs none: aOR 1.24 (95% CI 0.93–1.66) • Carbapenem vs none: aOR 1.24 (95% CI 1.02–1.51) *Secondary outcomes*: • aGVHD 3–4 carbapenem vs none (conditioning to GVHD onset): subHR 1.77 (95%CI 1.25–2.52) • GI GVHD carbapenem vs none: subHR 1.27 (95% CI 0.91–1.78)	Adjusted for: other antibiotics, age sex, race, transplant year, disease status, conditioning, graft source, donor, GVHD prophylaxis, performance score, CMV serostatus, ICU admission	Level 2b
Tanaka [120] (USA, 2020)	Single center retrospective cohort study	1214 patients including 607 children with allo-HSCT (age <18y) (part of a larger study)	Antibiotics with anaerobic activity (n = 491) (piperacillin-tazobactam or carbapenem), or no anaerobic activity (n = 723) (aztreonam, cefepime, or ceftazidime) (Day −7 to Day +28)	*Primary outcome: gut or liver aGVHD* • Anaerobic vs no anaerobic antibiotics: aHR 1.23 (95% CI 0.9–1.63) *Secondary outcomes*: • Skin aGVHD: aHR 1.10 (95% CI 0.86–1.41) • aGVHD mortality: 2.15 (95% CI 1.05–4.43) • cGVHD: 1.30 (95% CI 0.96–1.77) • cGVHD mortality: 0.90 (95% CI 0.42–1.92)	Adjusted for: HSCT year, age, sex, race, indication, HSCT donor, source, HLA matching, conditioning regimen intensity, and GVHD prophylaxis	Level 2b

(*Continued*)

Table 3. (Continued).

Author (country & year)	Study type	Population	Antibiotics investigated (HSCT exposure period)	Graft versus host disease data	Comments	Level of evidence*
Romick-Rosendale [121] (USA, 2018)	Single center prospective cohort study	42 HSCT for malignant and nonmalignant conditions (10 developed GVHD)	Anaerobic activity (piperacillin-tazobactam, meropenem, clindamycin, metronidazole)	• Short-chain fatty acids (acetate, butyrate, and propionate) decreased at day 7 and day 14 following HSCT compared with baseline • Total exposure to anaerobic antibiotics associated with low butyrate *(p* = 0.003) and propionate levels (*p* = 0.001) at + D14 • Butyrate (*p* = *0.0142*), propionate (*p* = *0*.0108) and acetate (*p* = *0.047*) levels were lower at +D14 in those with GVHD at any site compared with those that did not. • Formate levels at +D14 were not significantly altered in those patients who developed GVHD compared with those who did not	Indirect association between antibiotic exposure, short-chain fatty acids and GVHD. Luminal acetate levels were also lower with higher total exposure to antibiotics, although not to a significant degree	Level 2b
Simms-Waldrip [122] (USA, 2017)	Dual center prospective case-cohort pilot study	15 allo-HSCT for malignant and nonmalignant conditions with (n = 6) and without GVHD (n = 9)	Anaerobic activity (Piperacillin-tazobactam, carbapenem, clindamycin, metronidazole) vs no anaerobic activity (HSCT to day +43 or diagnosis of GVHD)	Primary outcome: grade 2–4 aGVHD or gut GVHD • Anaerobic vs non anaerobic DOT: OR 1.24 (95% CI 1.045–1.959) • Cumulative total antibiotic exposure DOT: OR 1.114 (95% CI 1.023–1.303)	Significant decline, up to 10-log fold, in gastrointestinal Clostridia in those with GVHD	Level 3b
Guthery [123] (USA, 2004)	Single center prospective pilot study with historical controls.	19 allo-HSCT compared with 83 historical controls	Enteral metronidazole 20 mg/kg/day (Day −14 to Day +35)	*Primary outcome: grade 2–4 aGHVD at day +100* • Metronidazole vs none: RR 0.36 (95% CI 0.13–0.997)	Adjusted for: age, unrelated donor status, GVHD prophylaxis 2 (10.5%) discontinued metronidazole due to vomiting	Level 3b

m, month; y, year; vs, versus; GVHD, graft versus host disease; *CIBMTR*, Center for International Blood and Marrow Transplant Research; IV, intravenous; a, acute; AMR, antimicrobial resistance; CDI, Clostridioides difficile infection; HR, Hazard ratio; CI, confidence interval; c, chronic.

* Grading per Oxford Center for Evidence-Based Medicine: Levels of Evidence (March 2009) [124].

pathogenesis. Commensal gut anaerobes, particularly Clostridiales, play an essential role in producing SCFAs such as acetate, butyrate, and propionate. Short-chain fatty acids, especially butyrate, have an important indirect immunomodulatory role by inducing differentiation of intestinal regulatory T cells (Tregs) and an increase in intestinal expression of forkhead box protein 3 (FOXP3). Tregs are responsible for immune homeostasis and may be key to moderating the severity of GVHD. Furthermore, butyrate supports the intestinal epithelial barrier as an epithelial energy source and by maintaining junction proteins [125]. In longitudinal profiling of the pediatric GM in allo-HSCT, preservation of obligate anaerobes, such as *Ruminococcaceae*, occurred in children who had earlier immune reconstitution, minimal aGVHD, and better overall survival [126].

Antibiotic prophylaxis has been reported to reduce the risk of GVHD in children; however, evidence is contradictory [115]. One proposed mechanism is that prophylaxis reduces episodes of FN, decreases cumulative antibiotic exposure, prevents disturbance of the GM and thereby reduces the risk of aGVHD. In 227 children receiving levofloxacin prophylaxis (day −2 to neutrophil recovery) mean antibiotic exposure was lower and grade 2–4 aGVHD was less compared with those not receiving prophylaxis [117]. Gut decontamination, using non-absorbable oral antibiotics to modify gut flora, has been associated with less severe aGVHD in several studies of children undergoing allo-HSCT [115,118,123]. In 19 children undergoing allo-HSCT, oral metronidazole (Day −14 to Day +35) was associated with less grade 2–4 aGHVD at day +100 when compared with children not receiving oral metronidazole therapy [123]. A retrospective study by Galazka *et al*. [118] found that gut decontamination with gentamicin, given with or without other agents from conditioning to neutrophil recovery, showed a statistically significant reduction in gastrointestinal GVHD and all grades of aGVHD. In contrast, they also showed that oral colistin and ciprofloxacin were associated with increased rates of any GVHD, albeit the study was limited by its study design and heterogenous use of decontamination antibiotics. Together, results from these pediatric studies should be interpreted with acknowledgment of contrasting data from larger studies in adults where antibiotic prophylaxis or gut decontamination is associated with increased rates or severity aGVHD [115].

Fecal microbiota transplantation (FMT) uses various methods to transfer the gut microbiota from healthy donors to patients with the goal of restoring GM homeostasis [127]. FMT has been used in patients undergoing HSCT to reduce colonization with resistant bacteria, treat recurrent *C difficile* infection, and to treat steroid-resistant or steroid-refractory GVHD [91,127,128]. One of the challenges is the multiple approaches to preparation of the donor sample, mode of delivery to the patient and timing of administration [128] as well as concern about the risk of bloodstream infection in immunocompromised patients with abnormal gut permeability [129]. Evidence for FMT efficacy and safety is mostly drawn from prospective cohort studies in adult patients. Overall FMT appears to have a low risk of immediate adverse effects, such as FMT-related infection, however the late effects of GM manipulation have not been established [127,128].

FMT may potentially ameliorate effects of GM dysbiosis and prevent GVHD. In a randomized, controlled trial fecal samples were stored prior to allo-HSCT and transplanted by retention enema after neutrophil engraftment in 14 adults. Compared with placebo, autologous FMT improved microbiota diversity and reestablished a GM resembling the pre-HSCT state [130]. DeFilipp *et al*. [131] conducted a similar study with FMT from unrelated donors delivered via oral capsule up to four weeks post neutrophil engraftment in 13 adults undergoing allo-HSCT. This approach was well tolerated and associated with reconstitution of GM diversity. There was particular benefit in the subset of patients who received broad-spectrum antibiotics before HSCT where this approach resulted in greater GM diversity at 30 days post-FMT. Fecal microbiota transplant may also be useful in the treatment of GVHD. In a meta-analysis of 242 predominately adult patients with GVHD, 161 (66.5%) had clinical improvement and 100 (41.3%) had complete response to FMT. Only two of the 23 included trials had a control group and all studies had a small sample size (≤52 participants) [128]. Evidence in children is limited to several case reports [128] and a small non-protocolized prospective single center study of FMT in children with gastrointestinal GVHD where partial or complete response was reported in the majority (6/7 children, 86%) [132]. Clinical trials are required to establish the safety of FMT in immunocompromised children and evaluate the efficacy of FMT to prevent and treat GVHD.

4. Expert opinion: conclusion and future directions

Fever, in particular the syndrome of febrile neutropenia, is a common complication in children with cancer or undergoing HSCT and drives repeated exposure to broad-spectrum antibiotics. Immediate administration of antibiotics is mostly a reflex response to neutropenic fever, however decisions around stopping antibiotics are much more complex. This is particularly true in children with high-risk FN where patients have repeated episodes of prolonged neutropenia and decision-making may be influenced by the patient's medical comorbidities, previous episodes of infection, and the perceived risk of adverse outcomes from serious infection. Decisions are also made in partnership with the young person and family who often have their own perspective on duration of antibiotics. When this is balanced against the paucity of evidence for duration of antibiotics in high-risk FN, it is understandable that the default is simply to 'continue.'

While this review highlights several research gaps, the known impacts of antibiotic exposure on the individual, the hospital and greater society are significant and should inform decisions about antibiotic use. Antibiotic exposure in children with cancer has a vast array of downstream effects including antimicrobial resistance, toxicity, marked modification of the GM, reduced efficacy of immunotherapy, decreased quality of life and significant financial cost. HSCT is the curative treatment of choice for many malignant and nonmalignant conditions, however exposure to broad-spectrum antibiotics may compromise engraftment, increase the risk of GVHD and add to transplant-related mortality. Ultimately, further research is needed to fully understand the interaction between antibiotics, the GM, response to cancer therapy and HSCT outcomes. It will also be important to describe any long-term influence antibiotics may have on the GM and outcomes for survivors of childhood cancer.

There is a delicate balance between the unchecked modification of the microbiome by broad-spectrum antibiotics and the targeted manipulation offered by selective antibiotic use, probiotics or microbial transplantation. While microbiota manipulation in patients with hematologic malignancies and in those undergoing HSCT seems like a panacea, to date, strategies such as FMT require further evidence. Ultimately, the immediate focus should be firmly on preserving the healthy GM by optimizing enteral nutrition and minimizing antibiotic exposure. Short course antibiotics are likely safe in many children with febrile neutropenia and present a valuable opportunity to reduce antibiotic exposure across the cancer treatment course. This review highlights the critical need for well-designed appropriately powered clinical trials to optimize the use of antibiotics in children with febrile neutropenia. Future research should examine the broader impacts of antibiotics, including health economic analysis and collection of quality-of-life data, and avoid solely focusing on measures such as 'days of antibiotic therapy.'

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